

SOLUTION

CASE STUDY 1 – SMART MEDICAL DIAGNOSIS SYSTEM

(a) Propositional Logic Representation

Let:

- **F** = Patient A has Fever
- **C** = Patient A has Cough
- **R** = Patient A has Rash
- **B** = Patient A has Breathlessness
- **Flu** = Patient A has Flu
- **Measles** = Patient A has Measles
- **Pneumonia** = Patient A has Pneumonia

Knowledge Base:

1. Fever AND Cough $\rightarrow$ Flu
 $F \wedge C \rightarrow \text{Flu}$
2. Fever AND Rash $\rightarrow$ Measles
 $F \wedge R \rightarrow \text{Measles}$
3. Cough AND Breathlessness $\rightarrow$ Pneumonia
 $C \wedge B \rightarrow \text{Pneumonia}$

Facts:

- **F = True**
- **C = True**
- **B = True**

(b) Forward Chaining

Step 1: Initial facts

Fever = True

Cough = True

Breathlessness = True

Step 2: Apply Rule 1

$F \wedge C \rightarrow \text{Flu}$

Since:

$F = \text{True}$ and $C = \text{True}$

Therefore:

Flu = True

Step 3: Apply Rule 2

$F \wedge R \rightarrow \text{Measles}$

Rash is not given as a fact.

Therefore:

Measles cannot be derived.

Step 4: Apply Rule 3

$C \wedge B \rightarrow \text{Pneumonia}$

Since:

$C = \text{True}$ and $B = \text{True}$

Therefore:

Pneumonia = True

Final diagnoses

- **Flu = True**
- **Pneumonia = True**
- **Measles = Not derivable**

(c) Backward Chaining for Pneumonia

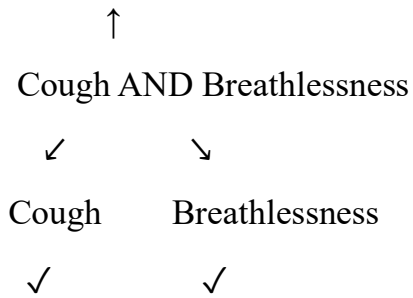
Goal: Pneumonia

Find a rule whose conclusion is Pneumonia:

Cough $\wedge$ Breathlessness $\rightarrow$ Pneumonia

Therefore, subgoals are:

Pneumonia



Both Cough and Breathlessness are facts.

Therefore:

Patient A has Pneumonia is proved.

CASE STUDY 2 – AUTONOMOUS TRAFFIC MANAGEMENT AGENT

Given FOL Rules

Rule 1:

$\forall x [\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)]$

Rule 2:

$\forall x [\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)]$

Rule 3:

$\forall x \forall y [\text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)]$

Facts

1. $\text{Vehicle}(\text{Ambulance})$
2. $\text{EmergencyType}(\text{Ambulance})$
3. $\text{Vehicle}(\text{CarA})$
4. $\text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Unification of Rule 1

Rule 1:

$\text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Step 1

Match:

Vehicle(x)

with:

Vehicle(Ambulance)

Therefore:

x = Ambulance

Substitution:

$\theta_1 = \{x/Ambulance\}$

Step 2

Check second condition:

EmergencyType(x)

becomes:

EmergencyType(Ambulance)

This is a given fact.

Therefore Rule 1 fires.

Step 3

Derived fact:

ClearPath(Ambulance)

Final MGU:

$\theta = \{x/Ambulance\}$

(b) Resolution Proof that Proceed(CarA) is True

Step 1 – Convert Rule 1 to CNF

Rule:

Vehicle(x) $\wedge$ EmergencyType(x) $\rightarrow$ ClearPath(x)

Eliminate implication:

$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Clause C1:

$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Step 2 – Rule 2

$\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$

This gives two clauses:

C2: $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

C3: $\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$

Step 3 – Rule 3

$\text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$

CNF:

C4: $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Step 4 – Negate the Goal

Goal:

Proceed(CarA)

For resolution refutation, add:

C5: $\neg \text{Proceed}(\text{CarA})$

Step 5 – Derive ClearPath(Ambulance)

Given:

$\text{Vehicle}(\text{Ambulance})$

$\text{EmergencyType}(\text{Ambulance})$

Using C1:

ClearPath(Ambulance)

Step 6 – Derive GreenSignal(Ambulance)

Using C2:

ClearPath(Ambulance)

gives:

GreenSignal(Ambulance)

Step 7 – Derive Proceed(CarA)

Given:

- Vehicle(CarA)
- Behind(CarA, Ambulance)
- GreenSignal(Ambulance)

Apply Rule 3:

Proceed(CarA)

But we also have:

¬Proceed(CarA)

Therefore:

Proceed(CarA) $\wedge$ ¬Proceed(CarA) $\rightarrow \square$

The empty clause $\square$ is obtained.

Conclusion

Therefore, by resolution:

$\therefore$ Proceed(CarA) is TRUE.

(c) Can the KB Handle Two Simultaneous Emergency Vehicles?

No, not completely with the current facts.

The rules themselves are general because they use variables, so they can apply to multiple emergency vehicles. However, the knowledge base currently contains emergency information only for:

Ambulance

There is no second emergency vehicle with a corresponding:

- Vehicle(...)
- EmergencyType(...)
- Behind(...)
- ClearPath(...)
- GreenSignal(...)

Additional facts required

For example:

Vehicle(FireTruck)

EmergencyType(FireTruck)

Behind(CarB, FireTruck)

If needed, traffic-control facts should also specify:

GreenSignal(FireTruck)

$\neg$ RedSignal(FireTruck)

Conclusion

The **FOL rules are reusable for multiple vehicles**, but additional facts about each emergency vehicle and its traffic conditions are required to reason about simultaneous emergencies.

CASE STUDY 3 – AGRICULTURAL EXPERT REASONING SYSTEM

Given KB

P1:

SoilDry $\rightarrow$ IrrigationNeeded

P2:

IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

P3:

$\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk

Facts:

- SoilDry = True

- CropWheat = True

(a) Convert to CNF

P1

Original:

SoilDry $\rightarrow$ IrrigationNeeded

Eliminate implication:

$\neg$ SoilDry $\vee$ IrrigationNeeded

Therefore:

C1 = $\neg$ SoilDry $\vee$ IrrigationNeeded

P2

Original:

IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

Eliminate implication:

$\neg$ (IrrigationNeeded $\wedge$ CropWheat) $\vee$ ApplyDripMethod

Using De Morgan's law:

$\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat $\vee$ ApplyDripMethod

Therefore:

C2 = $\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat $\vee$ ApplyDripMethod

P3

Original:

$\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk

Eliminate implication:

$\neg$ ($\neg$ ApplyDripMethod $\wedge$ CropWheat) $\vee$ CropAtRisk

Using De Morgan's law:

ApplyDripMethod $\vee$ $\neg$ CropWheat $\vee$ CropAtRisk

Therefore:

C3 = ApplyDripMethod $\vee$ $\neg$ CropWheat $\vee$ CropAtRisk

Facts in CNF

C4: SoilDry

C5: CropWheat

Final CNF

C1: $\neg$ SoilDry $\vee$ IrrigationNeeded

C2: $\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat $\vee$ ApplyDripMethod

C3: ApplyDripMethod $\vee$ $\neg$ CropWheat $\vee$ CropAtRisk

C4: SoilDry

C5: CropWheat

(b) Resolution Refutation – Prove ApplyDripMethod

Step 1 – Goal

Goal:

ApplyDripMethod

Negate the goal:

$\neg$ ApplyDripMethod

Add it to the KB.

Step 2 – Resolve C1 with SoilDry

C1:

$\neg$ SoilDry $\vee$ IrrigationNeeded

C4:

SoilDry

Resolution gives:

IrrigationNeeded

Step 3 – Resolve C2

C2:

$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

We have:

- IrrigationNeeded
- CropWheat

Therefore:

ApplyDripMethod

Step 4 – Resolve with negated goal

We have:

ApplyDripMethod

and:

$\neg \text{ApplyDripMethod}$

Therefore:

□

Resolution proof

SoilDry

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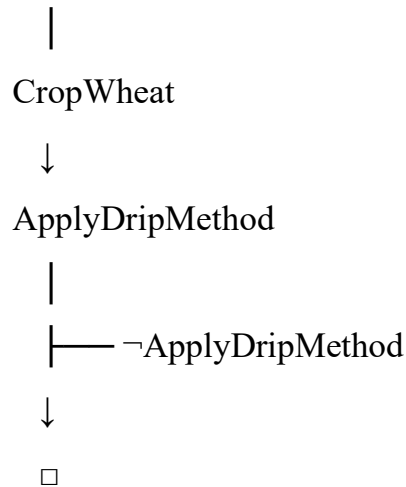
├── C1: $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

↓

IrrigationNeeded

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├── C2: $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$



Conclusion

ApplyDripMethod is logically entailed by the KB.

(c) Remove SoilDry from the KB

Initially:

SoilDry = True

CropWheat = True

P1:

SoilDry $\rightarrow$ IrrigationNeeded

Without SoilDry, we cannot derive:

IrrigationNeeded

Therefore P2 cannot fire:

IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

So:

ApplyDripMethod cannot be derived.

For P3:

$\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk

However, the absence of ApplyDripMethod does **not automatically mean** $\neg$ ApplyDripMethod.

Therefore, under standard propositional logic:

CropAtRisk also cannot be logically concluded unless $\neg \text{ApplyDripMethod}$ is explicitly given.

CASE STUDY 4 – WUMPUS WORLD KNOWLEDGE AGENT

Given Percepts

- Stench at [1,2]
- Breeze at [1,1]
- Glitter at [2,2]

Rules

1. $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to } [1,2]$
2. $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to } [1,1]$
3. $\text{Glitter}[x,y] \rightarrow \text{Gold is at } [x,y]$
4. $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

(a) Belief State Using Modus Ponens

Step 1 – Stench

Fact:

Stench[1,2]

Rule:

Stench[1,2] $\rightarrow$ Wumpus is adjacent to [1,2]

Therefore:

Wumpus is adjacent to [1,2].

The adjacent cells are:

- [1,1]
- [1,3]
- [2,2]

Therefore possible Wumpus locations:

{[1,1], [1,3], [2,2]}

Step 2 – Breeze

Fact:

Breeze[1,1]

Rule:

Breeze[1,1] $\rightarrow$ Pit is adjacent to [1,1]

Adjacent cells are:

- [1,2]
- [2,1]

Therefore possible Pit locations:

{[1,2], [2,1]}

Step 3 – Glitter

Fact:

Glitter[2,2]

Rule:

Glitter[x,y] $\rightarrow$ Gold[x,y]

Substitute:

$x = 2, y = 2$

Therefore:

Gold[2,2]

Derived knowledge

Wumpus $\in \{[1,1], [1,3], [2,2]\}$

Pit $\in \{[1,2], [2,1]\}$

Gold = [2,2]

(b) Forward Chaining

Step 1

Start with:

Stench[1,2]

Apply Rule 1:

Wumpus adjacent to [1,2]

Possible locations:

[1,1], [1,3], [2,2]

Step 2

Start with:

Breeze[1,1]

Apply Rule 2:

Pit adjacent to [1,1]

Possible locations:

[1,2], [2,1]

Step 3

Start with:

Glitter[2,2]

Apply Rule 3:

Gold[2,2]

Forward-chaining result

Stench[1,2]



Wumpus adjacent to [1,2]



Possible Wumpus:

[1,1], [1,3], [2,2]

Breeze[1,1]



Pit adjacent to [1,1]



Possible Pit:

[1,2], [2,1]

Glitter[2,2]



Gold[2,2]

(c) Is the Path [1,1] → [1,2] → [2,2] Safe?

Cell [1,1]

We know there is a **Breeze** at [1,1].

This indicates that a pit may be adjacent to [1,1].

Possible pit locations include:

- [1,2]
- [2,1]

Therefore, [1,1] is **not proven safe**.

Cell [1,2]

There is a **Stench** at [1,2].

This indicates a possible Wumpus in an adjacent cell.

Also, [1,2] itself is a possible location of the pit because of the Breeze at [1,1].

Therefore, **[1,2] cannot be proven safe.**

Cell [2,2]

There is **Glitter** at [2,2], so:

Gold[2,2]

However, the rules do not establish:

$\neg$ Wumpus[2,2]

or

$\neg$ Pit[2,2]

Therefore Rule 4 cannot prove:

Safe[2,2]

Final Conclusion

The path:

[1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2]

is **NOT logically guaranteed to be safe.**

The agent should **not assume safety merely because no hazard is directly observed.**

The knowledge base needs additional negative facts such as:

$\neg$ Wumpus[2,2]

$\neg$ Pit[2,2]

and corresponding information for the other cells before Rule 4 can establish safety.

Final answer:

The agent can identify [2,2] as the location of the gold, but it cannot prove that the complete path is safe with the given knowledge base.